

Concept Review

Stepper Motors

Why Steppers?

A stepper motor is a brushless, synchronous electric motor that moves in discrete steps. Unlike DC or BLDC motors, which rotate continuously when powered, a stepper motor divides a full rotation into a number of equal steps. When driven with the right sequence of electrical pulses, it turns by a small amount, making it ideal for precise position and speed control without requiring feedback sensors.

Stepper motors are commonly used in applications like 3D printers, CNC machines, and robotics where accuracy and repeatability are more important than raw speed or torque.

1. Introduction

For many mechatronic applications, it is necessary to be able to control the position of an actuator with precision. In contrast to the continuous rotation of a brushed or brushless DC motor, a stepper motor rotates the armature in discrete steps. It uses fixed coil spacing to the motor to be firmly held in a known position and moved with reasonable certainty to a new position by steps of predefined size. The number of steps it takes to complete a full revolution determines the stepper motor's step size. As the motor is only stepping between known distinct positions, it is possible to command it to hold a certain position without requiring any feedback position sensors. This approach is called open-loop control and is relatively simple to implement. To perform the same position control task with brushed or brushless DC motors requires sensor data feedback and is called closed-loop control.

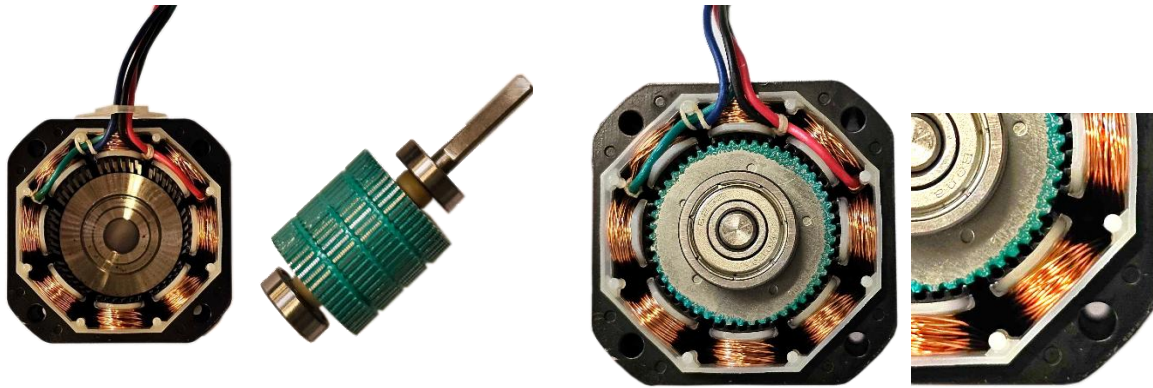
2. Basic Construction and Drive

A stepper motor consists of two parts, the rotor and the stator.

- The rotor, as the name suggests, is the rotating part which is typically a permanent magnet or a toothed iron core that reacts to the magnetic fields generated by the stator.
- The stator contains multiple windings arranged in phases, which are energized in a specific sequence to produce a rotating magnetic field. As the field rotates, the rotor aligns with it, stepping forward with each pulse of current.

The number of steps per revolution is determined by the internal design of the motor—common stepper motors have a step angle of 1.8° , resulting in 200 steps per full rotation.

The stepper motor that is provided with the Mechatronic Actuators Trainer is a hybrid stepper motor that has a permanent magnet toothed rotor and a toothed stator. Figure 1 (a) shows the two different pieces, the stator and the rotor and Figure 1 (b) shows the inside of an assembled stepper motor and how some teeth from the rotor align with the stator and some don't.



(a). Stator with 8 coils and Rotor

(b) Assembled Stepper and teeth alignment

Figure 1. Inside of a Stepper Motor

The stepper motor that is provided with the Mechatronic Actuators Trainer has 8 poles that are activated by winding A and B. When activating winding A, 4 of the poles will be magnetized, 2 of them with north polarity and 2 with south polarity. As shown in Figure 2, rotor teeth are aligned with the A windings and not with the B, as the sequence of steps enables B next, the rotor will align with B, therefore generating the movement as the alignment happens, doing this repeatedly is what generates stepper motor rotation.

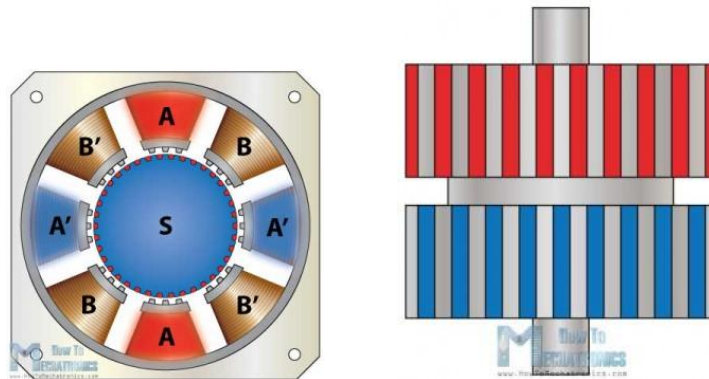


Figure 2. Toothed Hybrid Stepper and Side view of Rotor

The way these hybrid steppers work is by having half of the rotor be a north pole and half of it being a south pole as shown in Figure 2. The rotor in the provided stepper has 50 teeth, which is the most common for hybrid stepper motors. If the stator had 50 teeth, the magnets would always align, therefore, not allowing for any rotation, for that reason, the stator tends to have 2 fewer teeth.

At each step, 1 pair of stator teeth will be fully aligned, 2 will be half aligned and 1 will be unaligned as shown in Figure 3. When using Figure 2 as a reference, we can see that when coil A is energized, The tooth are fully aligned with the teeth in the stator A, since it acts as a north pole, and therefore attracting the teeth in the south pole of the

rotor but fully unaligned with A' since the south side of the magnet in the stator is being repelled by the south pole generated by A'.

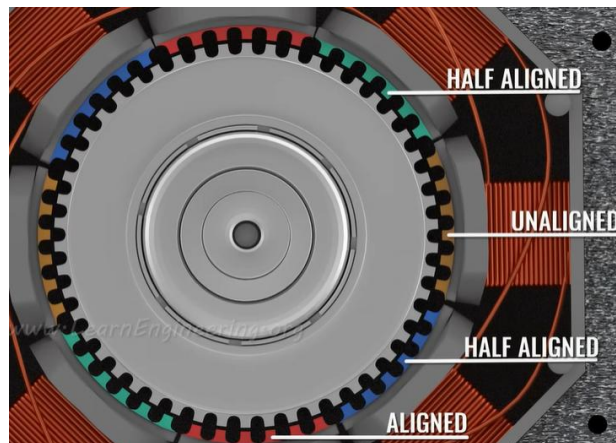


Figure 3. Rotor and Stator Teeth Alignment

When coil B is energized, the rotor rotates a small angle to align with the teeth that were half aligned in the previous step, since that is the new north/south pole. As coil A and B are energized, they attract and repel the rotor's magnetic field constantly to generate the rotation. In the case of a stepper with 50 teeth, it takes 4 steps (A,B,A' and B') for the teeth to align with the teeth next to the starting point, therefore, every step taken moves the rotor by 1.8 degrees.

a. Unipolar/Bipolar Motors

The two basic winding configurations for stepper motors are unipolar and bipolar windings (Figure 4).

- Unipolar motors have 6 wires, 4 for the motor windings and a center tap on each phase. Therefore, changing the polarity only requires choosing the other section of the winding. This type of commutation can be achieved by a single transistor for each winding.
- Bipolar motors only require 4 wires, 2 for each winding. In the case of a bipolar stepper motor, the stator has two windings, each requiring bidirectional current. Unlike unipolar motors, which use center-tapped windings to simplify control, bipolar stepper motors offer higher torque and better efficiency by utilizing the entire winding during operation. However, because each coil must be energized in both directions, an H-bridge circuit is typically used to control the current flow through the windings. The circuit differences can be seen in Figure 4.

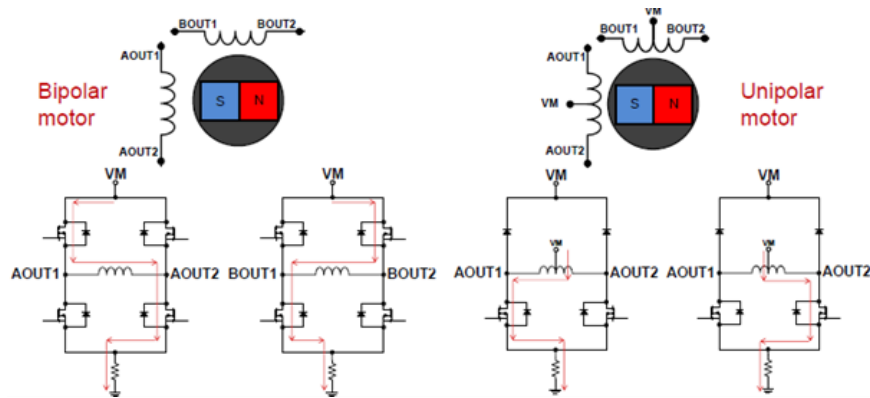


Figure 4. Unipolar vs Bipolar Motor

3. Stepper Excitation Modes

There are several standard ways to drive the stepper motor. These are called stepping modes, which affect the resolution, torque, and smoothness of the motor. We will use a simplified diagram of a permanent magnet stepper motor to simplify the descriptions; however, the working principle remains for the hybrid stepper motor, the only difference lies in the number of steps per full rotation.

a. Wave Drive

In wave drive (or single-coil excitation), only one coil is energized at a time. This method minimizes power consumption but provides lower torque since only one winding is active per step. As shown in Figure 5, when coil A is energized, the south pole of the rotor magnet aligns with it. When coil B is energized next, the magnet will rotate clockwise, aligning the south pole with B. In the case of this example, it will take 4 steps to do a full cycle rotation, in other motors, look at the step size to know how many steps it will take to do a full 360 rotation.

It is important to note that when we refer to energizing A' or B', it means applying current in the opposite direction compared to A or B. This reverses the magnetic polarity of the winding, effectively making A' or B' behave like a north pole when A or B would act as a south pole.

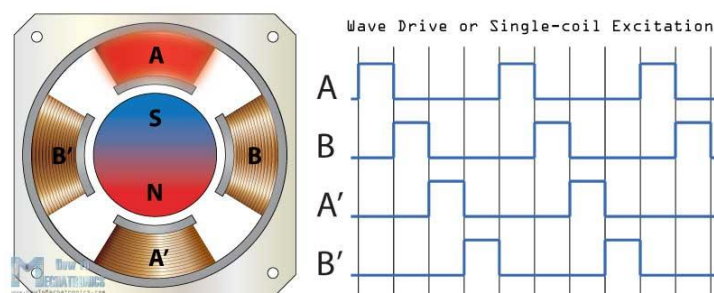


Figure 5. Stepper Wave Drive

b. Full Step Drive

To achieve the rated torque of the motor, a full step drive is used. In this excitation mode, two neighboring phases are active at any given time, resulting in the maximum rated torque and the rotor aligning with a position between the active phases (Figure 6). However, resolution is the same, and it takes the same number of steps to do a full rotation as in wave drive but doing it with significantly higher torque.

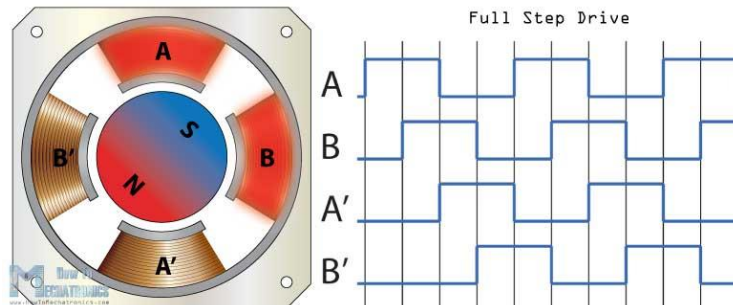


Figure 6. Stepper Full Step Drive

c. Half Step Drive

Another excitation mode which combines the previous two modes is the half-step drive. Here, the motor alternates between one and two active coils, resulting in double the angular resolution of wave and full step drives. Which means it needs double the number of steps to complete one full rotation. In this mode, the holding torque varies with each half step depending on whether one or two phases are active.

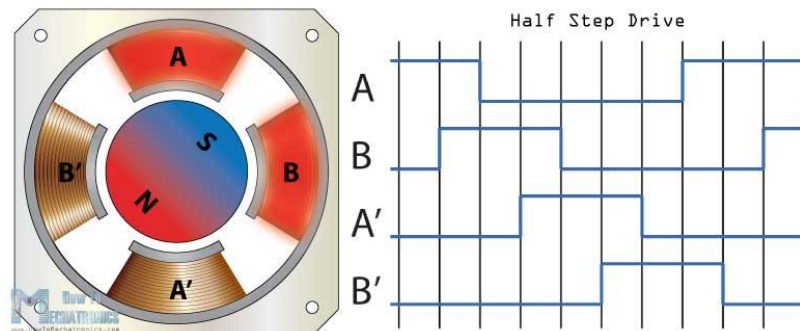


Figure 7. Stepper Half Step Drive

d. Microstepping

Lastly, a stepper motor can be excited using micro-stepping. In this excitation mode, the current in each of the phases approximate sinusoids, allowing for very smooth motor operation. The most common way this mode is implemented is sine-cosine micro-stepping and is depicted in Figure 8. In contrast to the other excitation modes described above, micro-stepping requires a more complex amplifier circuitry; however, properly implemented, it allows extremely fine position control. The torque achieved is highest in a holding position and decreases with an increase in speed.

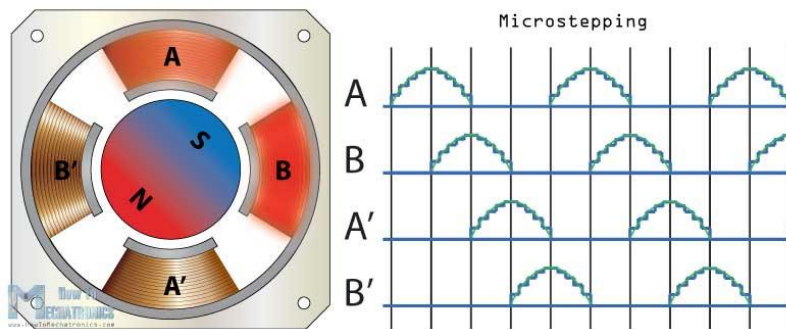


Figure 8. Microstepping

4. Datasheet And Considerations

The motor's datasheet summarizes its key electrical and mechanical characteristics and are essential when selecting a motor for a project or application.

a. Understanding a Stepper Motor Datasheet

When selecting a DC motor, there are certain parts of the datasheet that might be important when looking for motors for a project. This section will mostly focus on the electrical information, however, note that there will also be mechanical information about the motor, usually in the form of a mechanical drawing. This information is important for deciding if a motor will physically fit in a required space, or mount effectively. There may also be information on the total weight of the actuator, which may be important for applications such as serial manipulators, or model airplane control surface actuators.

When selecting a stepper motor, there are several pieces of specification data which will be of particular interest. Figure 9 shows two excerpts from the datasheet for the stepper motor provided with the actuators trainer.

● Specifications

■ Bi-polar

Series & Length	Model Number	Holding Torque		Rated Current	Resistance per Phase	Inductance per Phase	Detent Torque		Rotor Inertia	
		N.cm	oz-in	A	ohm	mH	N.cm	oz-in	g.cm ²	oz-in ²
PHB42S48 48mm	PHB42S48-401	52	72.8	1.3	3.2	5.5	2.6	3.64	68	0.374
	PHB42S48-402	52	72.8	1.7	1.8	3.2	2.6	3.64	68	0.374
	PHB42S48-403	52	72.8	2.3	1.2	1.6	2.6	3.64	68	0.374

Step Angle	1.8°
Step Angle Accuracy	±5% (full step, no load)

Figure 9. Stepper motor specifications

This is a fairly standard set of specifications for a stepper motor; however, it lacks some useful information, which will be outlined below. Key information which should be provided for a stepper motor will be:

- | | |
|----------------|--------------------|
| 1. Step Angle | 5. Holding torque |
| 2. Coil wiring | 6. Pull-in torque |
| 3. Voltage | 7. Pull-out torque |
| 4. Current | 8. Detent torque |

Several factors will likely be important for deciding whether or not a motor is appropriate for your application.

i. Rated Voltage

This is the voltage at which the motor is designed to run, and at which the motor will have been tested for other specifications. If the motor voltage is significantly greater than that of the power supply driving it, the system will be unable to produce the rated torque. If the supply voltage exceeds the rated voltage, there is a risk of damaging the motor hardware.

ii. Rated Current

This is the maximum current each phase can handle continuously without overheating. The peak current is drawn when the coil activation changes. However, the current drawn for the stepper will be relatively consistent as long as a coil is energized. Since stepper motors draw current even at standstill, exceeding the rated current can cause excessive heat buildup. Undersupplying current reduces available torque and may lead to skipped steps under load.

iii. Max/Holding Torque

Holding torque is the maximum torque the motor can resist when energized but stationary. This is typically the highest torque rating for a stepper and is important when the motor must maintain position under load. If the applied load exceeds holding torque, the motor will skip steps, resulting in loss of position. This results in not having a way to know the actual position of the motor if it moves in this way.

iv. Pull-in/Pull-out Torque

These are specific torque values that define the behavior of a stepper motor in motion. Pull-in torque is the maximum torque under which the stepper will start without losing a step. Pull-out torque is the maximum torque that can be applied to a stepper motor in motion before it begins to skip steps. Both values are inversely related to the stepping rate commanded, they decrease as stepping rate increases.

v. Detent Torque

This is the amount of torque that an unpowered stepper motor will apply due to internal magnetic forces in reaction to an external rotation. This is the torque which causes the “cogging” you feel when rotating a stepper motor manually.

vi. Motor Speed

Stepper motors produce less torque at higher speeds. Manufacturers often provide a speed-torque curve that shows how torque declines as stepping speed increases. This curve is essential for selecting a motor that meets both speed and load requirements.

For stepper motors, the maximum speed will be a function of the desired pull-out torque. This torque curve may be given in a plot such as that shown in Figure 10.

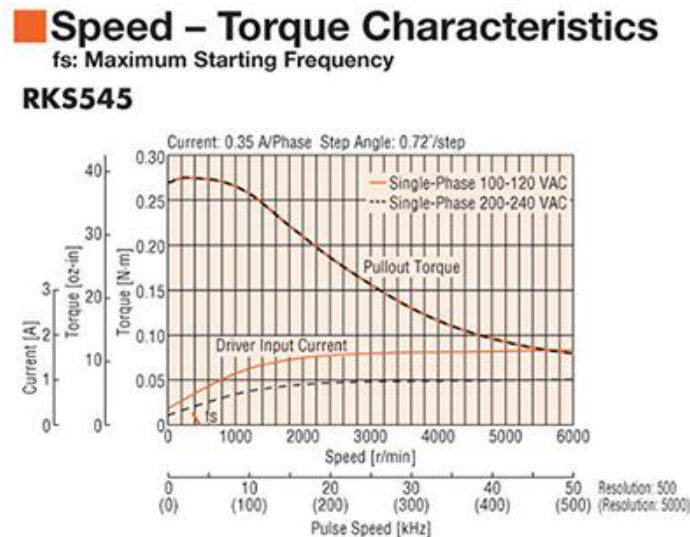


Figure 10. Speed-torque curve for a stepper motor

This plot will show the maximum speed the motor is capable of achieving under a given torque load.

b. Design Decisions

i. Cost

Stepper motors are generally cost-effective, especially for moderate- to high-torque applications. However, the overall cost of using a stepper motor includes not only the motor itself but also the required driver electronics, which are more complex than simple brushed motor controllers. These additional components can raise the total system cost, particularly for low-volume designs.

ii. Efficiency

Stepper motors are not particularly efficient in terms of power-to-work conversion. They consume constant current even when holding position (high current) or under no load, which leads to continuous power dissipation and heat generation. Many stepper motors are designed to operate safely at elevated temperatures, often exceeding 100 °C, but thermal management may still be necessary in tightly enclosed systems.

iii. Ease of Use

Controlling a stepper motor requires generating specific and precisely timed excitation signals, often through a microcontroller or dedicated driver IC. This makes setup more complex than something like a brushed DC motor. However, modern stepper drivers can do much of this complexity, providing simple interfaces such as direction and step input that make integration easier, however, this could increase cost.

iv. Lifespan

Stepper motors tend to have long operational lives due to their simple, brushless construction and lack of internal feedback components.

v. Range of Motion

Stepper motors offer continuous 360° rotation and are well-suited for applications requiring unlimited or repeated motion in either direction.

vi. Position Control Accuracy

Stepper motors offer precise position control through open-loop stepping. With techniques like microstepping, the resolution can be increased significantly, however, it causes a significant loss of holding torque. However, accuracy can degrade if steps are missed due to excessive load or inertia, since there is no built-in feedback.

c. Comparison To Other Motors

Pros:

- Do not require sensors to detect position
- Good torque at low speeds
- Great at holding positions
- Long Lifespan
- Repeatable movements. Step error is non-cumulative.

Cons:

- Can miss steps, no feedback
- High current draw, which makes them warm
- Lower torque at higher speeds

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